***Female Reproductive System & Uterine Cycle*: Professional Examination**

1. A 28-year-old female with a regular 28-day menstrual cycle is undergoing clinical monitoring. On approximately which day of her uterine cycle does ovulation typically occur?
a. Day 1
b. Day 5
c. Day 7
d. Day 11
e. Day 14
2. During a histological review of an ovarian biopsy, a tissue sample reveals a thick, translucent, acellular glycoprotein layer surrounding a primary oocyte. This specific layer is the:
a. Corona radiata
b. Theca interna
c. Theca externa
d. Zona pellucida
e. Antrum
3. At which precise stage of meiosis is the primary oocyte arrested within the primordial follicle from the time of birth until just before ovulation occurs?
a. Metaphase of meiosis I
b. Anaphase of meiosis II
c. Prophase of meiosis II
d. Prophase of meiosis I (Dictyotene stage)
e. Telophase of meiosis I
4. A patient undergoing an endometrial biopsy is found to have highly coiled, tortuous uterine glands actively secreting glycogen-rich fluid. This histological profile is characteristic of which phase of the uterine cycle?
a. Proliferative phase
b. Menstrual phase
c. Ischemic phase
d. Follicular phase
e. Secretory phase
5. During a routine pelvic examination, a clinician explains that the standard anatomical site where fertilization of the human oocyte by a spermatozoon occurs is the:
a. Uterine cavity
b. Fundus of the uterus
c. Cervical canal
d. Ampulla of the uterine tube
e. Infundibulum
6. Which hormone is directly responsible for inducing the rapid proliferation of the endometrial functional layer and the repair of the uterine wall following menstruation?
a. Progesterone
b. Luteinizing hormone
c. Follicle-stimulating hormone

d. Estrogen
e. Human chorionic gonadotropin

7. Following ovulation, the remnants of the ruptured Graafian follicle collapse and reorganize into a temporary endocrine structure known as the:
a. Corpus albicans
b. Corpus atreticum
c. Corpus luteum
d. Theca lutein
e. Granulosa lutein

8. A 35-year-old female tracks her menstrual cycle and notices a sudden surge in which hormone approximately 24 to 36 hours prior to the release of the secondary oocyte?
a. Progesterone
b. Estrogen
c. Follicle-stimulating hormone
d. Human chorionic gonadotropin
e. Luteinizing hormone

9. The functional layer of the endometrium undergoes extensive sloughing and necrosis during menstruation. This rapid shedding is triggered directly by a sudden withdrawal of:
a. Estrogen and Progesterone
b. Luteinizing hormone
c. Follicle-stimulating hormone
d. Oxytocin
e. Prolactin

10. At the time of ovulation, the secondary oocyte is discharged from the mature follicle. At which specific stage of meiosis is this cell arrested when it enters the uterine tube?
a. Prophase I
b. Metaphase II
c. Anaphase I
d. Telophase II
e. Metaphase I

11. Which cells of the developing ovarian follicle express luteinizing hormone (LH) receptors and are responsible for synthesizing androgens from cholesterol?
a. Theca interna cells
b. Granulosa cells
c. Corona radiata cells
d. Cumulus oophorus cells
e. Zona pellucida cells

12. Following the degeneration of the corpus luteum due to a lack of pregnancy, the endocrine tissue undergoes fibrotic changes and forms a white connective tissue scar called the:
a. Corpus luteum of pregnancy
b. Corpus albicans
c. Corpus atreticum
d. Macula lutea
e. Tunica albuginea

13. A patient is found to have an imperforate hymen blocking the opening of the female copulatory organ. This distensible muscular tube is the:
a. Uterus
b. Cervix
c. Vagina
d. Oviduct
e. Clitoris
14. The dynamic layer of the uterus that undergoes cyclic structural transformations, thickening, and shedding under the influence of ovarian steroid hormones is the:
a. Myometrium
b. Perimetrium
c. Stratum basale
d. Endometrium (Stratum funktionale)
e. Parametrium
15. Under the influence of follicle-stimulating hormone (FSH), granulosa cells convert thecal androgens into estrogens using which specific enzyme?
a. 5-alpha reductase
b. Aromastase
c. 21-hydroxylase
d. Protease
e. Hyaluronidase
16. A laboratory model replicates a mutation that completely blocks the expression of progesterone receptors within the endometrial stroma. What conceptual defect will be observed in the uterine cycle?
a. The proliferative phase will fail to initiate
b. The functional layer will fail to transition into the secretory phase, preventing proper preparation for implantation
c. Ovulation will happen twice in every cycle
d. Granulosa cells will fail to synthesize any estrogen
e. The primary oocyte will fail to finish its first mitotic division
17. If a pharmacological agent systematically prevents the local production of prostaglandins within the spiral arteries of the endometrium at the end of the secretory phase, what is the downstream mechanical effect on the uterine cycle?
a. The proliferative phase will become permanently accelerated
b. The ischemic phase and subsequent vasoconstriction will be blocked, severely disrupting normal menstruation
c. The corpus luteum will turn directly into a choriocarcinoma
d. Estrogen will be forced to undergo conversion into testosterone
e. The zona pellucida will dissolve prematurely inside the ovary
18. A clinical evaluation reveals that a patient possesses an abnormally low level of circulating estrogen during the mid-follicular phase, yet her FSH levels are elevated. What is the expected conceptual outcome regarding the selection of a dominant follicle?
a. Dozens of follicles will undergo simultaneous ovulation
b. The granulosa cells are failing to respond to FSH or lack adequate aromatase activity, preventing the selection and maturation of a dominant follicle
c. The endometrium will immediately enter the ischemic phase
d. The primary oocyte will instantly complete meiosis II

e. The vaginal epithelium will transform into stratified columnar cells

19. Why does the administration of a continuous, non-pulsatile high-dose GnRH agonist paradoxically cause a complete shutdown of the ovarian and uterine cycles in patients with endometriosis?

a. It stimulates the overproduction of progesterone from the adrenal cortex
b. It mechanically tears the basement membrane of the primordial follicles
c. It down-regulates pituitary GnRH receptors, suppressing LH and FSH secretion and leaving the ovaries completely unstimulated
d. It causes the immediate fibrotic regression of the entire stratum basale
e. It forces the secondary oocyte to remain trapped in metaphase I

20. A 29-year-old female with an endocrine disorder exhibits a complete absence of the mid-cycle LH surge, though her estrogen levels rise normally. Conceptually, what is the direct developmental outcome of this specific hormonal defect?

a. The endometrium will turn directly into a decidua
b. The primary oocyte will complete meiosis I but fail to enter the uterine tube
c. The secretory phase will trigger a massive hemorrhage
d. Ovulation will fail to occur, and the primary oocyte will remain trapped within an unruptured follicle
e. The connecting stalk will fail to migrate ventrally

21. If the spiral arteries of the endometrium fail to regress or undergo spasm when progesterone levels decline at day 26, how will the physical mechanism of menstruation be altered?

a. Sloughing of the functional layer will be clean and rapid without blood loss
b. The ischemic necrosis of the stratum funktionale will be impaired, causing irregular, prolonged, or absent shedding
c. The acrosomal cap will prematurely degrade within the cervix
d. The stratum basale will be shed instead of the stratum funktionale
e. The secondary yolk sac will completely fail to undergo constriction

22. In a conceptual scenario where the cells of the theca interna completely lack functional LH receptors, what is the downstream pathophysiological effect on the ovarian cycle?

a. Granulosa cells will overproduce progesterone during week 1
b. Granulosa cells will lack the necessary androgen precursors to synthesize estrogen, halting follicular maturation
c. The oocyte will automatically expel two polar bodies simultaneously
d. Semen volume will drop within the uterine tube
e. The corpus albicans will enter a state of hypersecretion

23. If an experimental drug selectively blocks the secretion of alkaline, watery mucus by the endocervical glands during the late proliferative phase, what biological barrier will face arriving spermatozoa?

a. They will fail to undergo compaction in the ampulla
b. They will be unable to penetrate the thick, acidic, and highly viscous cervical mucus, preventing entry into the uterine cavity
c. Their flagella will automatically grow to twice their normal length
d. They will become completely immune to the uterine immune cells
e. They will automatically stimulate the secondary oocyte to finish meiosis I prematurely

24. During a research study on the uterine cycle, the basal artery branches of the uterine vasculature are selectively ligated. Despite this, the stratum funktionale is successfully regenerated during subsequent cycles. This occurs because:
a. The stratum funktionale is nourished entirely by amniotic fluid
b. The straight arteries supplying the stratum basale remain intact, preserving the regenerative cellular stem pool
c. The spiral arteries can synthesize blood cells locally from the epithelium
d. The endometrial glands secrete blood directly into the cavity
e. The myometrium undergoes complete metaplasia into vascular tissue
25. A patient with a rare genetic mutation produces normal levels of estrogen but completely lacks the ability to execute positive feedback on the anterior pituitary gland. What structural and functional alteration will be observed?
a. Complete absence of primordial follicles at birth
b. A total absence of the mid-cycle LH surge, preventing ovulation despite adequate follicular growth
c. Massive hypertrophy of the tunica albuginea
d. Complete failure of the uterine tube to develop cilia
e. Metaplasia of the endometrium into stratified squamous epithelium
26. Conceptually, why does the presence of a functioning syncytiotrophoblast layer during week 3 prevent the onset of a new menstrual and uterine cycle?
a. It directly destroys the tight junctions of the endometrial capillaries
b. It secretes hCG to maintain the corpus luteum's progesterone production, preventing the ischemic phase of the endometrium
c. It forces the stratum basale to transform into a primary yolk sac
d. It causes the cervical glands to secrete a highly acidic fluid that kills maternal tissue
e. It blocks the pituitary from organizing its regular homeostatic rhythm
27. If a secondary oocyte is fertilized but the resulting blastocyst completely fails to produce human chorionic gonadotropin (hCG), what is the inevitable clinical consequence?
a. The embryo will develop two identical heads
b. The corpus luteum will regress, causing progesterone levels to plunge and the endometrium to slough off, terminating the early pregnancy
c. The functional layer will become twice as thick as normal
d. The embryo will fail to express androgen receptors on its surface
e. The blastocyst will implant exclusively in the liver parenchyma
28. A 32-year-old female exhibits a chronic, severe blockage of her bilateral oviducts at the junction of the ampulla and isthmus. Anatomically and conceptually, this obstruction prevents which two biological entities from meeting?
a. The maternal blood and the fetal trophoblast layers
b. The ascending spermatozoa and the ovulated secondary oocyte
c. The primitive streak cells and the notochordal process
d. The primary yolk sac and the amniotic fluid
e. The epiblast and the hypoblast layers
29. If the tight junctions forming the blood-follicle barrier are experimentally breached during the antral follicle stage, what is the conceptual consequence inside the ovary?
a. The primary oocyte will accelerate its mitotic divisions
b. Systemic immune cells and proteins can enter the follicular fluid, altering the microenvironment and impairing oocyte maturation

c. The granulosa cells will safely enter the bloodstream and travel to the spleen
d. The theca externa will undergo immediate apoptosis
e. The endometrium will undergo compensatory hypertrophy

30. What is the operational functional significance of the highly acidic pH characteristic of the normal vaginal fluid during the reproductive years?

a. To induce the regression of primitive streak remnants
b. To trigger the final condensation of the spermatid nucleus
c. To maximize the structural rigidity of the pelvic walls
d. To inhibit the growth of pathogenic microorganisms, maintaining an antimicrobial barrier despite the regular cycle changes
e. To activate the countercurrent heat exchange inside the uterine tube

HMHDental

Answer Key

Q 1-8	Q 9-16	Q 17-24	Q 25-30
e	b	b	b
d	b	a	b
d	a	e	b
e	c	a	b
d	b	d	d
d	c	a	d
c	e	d	
e	b	b	